

Morgan Bleasdale

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Education

Imperial College London

2025 – 2029

MEng Electrical and Electronic Engineering

- On track for First-Class Honours (graduation 2029)
- Awarded national UKESF Semiconductor Talent Award for potential in the semiconductor industry.

Kings Monkton School

2021 – 2025

*A Levels: A*A*A*A in Mathematics, Further Mathematics, Physics and Computer Science*

*GCSEs: 11 A*s, 1 A and 1 Distinction*

- Awarded Welsh Engineering Student of the Year Runner-Up (2024) by Engineering Education Scheme Wales.
- Awarded Principal's Prize (2025), Mathematics Prize (2023, 2025), Leadership Prize (2023, 2024).
- Obtained Gold CREST Award for leading a team to design the power system for a sustainable community.

Experience

Vishay Intertechnology, Inc | Newport, UK

July 2026 – September 2026

Summer Intern - Semiconductors

- Analysed 344,000+ production data points in MOSFET testing to implement non-existing control limits, preventing out-of-spec parts being shipped undetected; presented to Process Change Board for approval.
- Designed a data-driven process recipe selection framework for Active Area Deposition to reduce recurring false-positive out-of-specification holds caused by incorrect recipe selection.
- Reduced part qualification request completion time by 80% by developing a Python tool to compile part specification and qualification test data and automatically populate parameterised forms.
- Investigated a silicon staining defect affecting ~3% of production wafers by analysing focused ion beam failure analysis images and comparing to fab tool data; presented potential root causes to the process team.
- Developed manufacturing process flow documentation for silicon carbide (SiC) diode.

National Aeronautics and Space Administration (NASA) | Florida, USA

July 2024 – August 2025

VP of Engineering, International Space Settlement Design Competition

- Selected twice to represent the UK (top 12 of 1000+ applicants), leading team of 60+ engineering students to design a space station. 2024: Winner (Lunar data centre), 2025: Runner-up (Venusian Mining Settlement).
- Led overall technical design, managing 4 engineering departments with the systems engineering V-Model and a requirements traceability matrix to ensure the 50-slide proposal was cohesive and complied with RFP.

Recent Projects

FPGA Mixed Signal Oscilloscope and Signal Generator (SystemVerilog)

August 2026 – September 2026

Synthesised and deployed a SystemVerilog RTL oscilloscope/signal generator to Intel Cyclone IV FPGA using Quartus.

- Interfaced a 2 MSps parallel ADC as probe input (Nyquist limit of 1 MHz), buffering samples in FPGA-side BRAM with edge-triggered capture (adjustable threshold) to isolate waveform around a trigger event.
- Decimated waveform on-chip, reducing data rate to ensure compatibility with UART/USB bandwidth limits.
- Implemented Direct Digital Synthesis for signal generation using a BRAM-stored waveform lookup table, with frequency set via a phase-increment register and amplitude set via a gain register.
- Streamed decimated samples to laptop over UART/USB and built a web-based UI (Next.js) to render trace.

RISC CPU with custom ISA (SystemVerilog)

June 2026 – July 2026

16-bit Harvard-architecture RISC CPU in SystemVerilog, inspired by the design principles of ARM architecture.

- Implemented a single-cycle data path with an 8-operation ALU and NZCV flag registers, where flag states are updated in one cycle and consumed by the next cycle to support carry-chained operations and jumps.
- Developed flag-based conditional jumping (16 conditions, derived from NZCV status flags set by ALU operations), enabling structured control flow such as loops, conditionals, and subroutine calls/returns.
- Built load/store architecture with dedicated memory instructions, interfacing an 8x16-bit register file with data RAM and PC register with instruction ROM, supporting register-offset or direct addressing.
- Verified RTL using SystemVerilog testbenches in Icarus Verilog with a 100% pass rate; demonstrated functionality with a custom assembly program to perform a multi-word 16x16-bit software multiplication.

Multi-sensor remote-controlled Wi-Fi Lunar Rover

May 2026 – June 2026

Built an RC rover that surveys a lunar environment, detecting RF, IR, ultrasonic signals, and magnetic fields.

- Built the radio subsystem with a tuned LC coil antenna, inverting amplifier, envelope detector, and hysteresis comparator to demodulate an 89 kHz ASK-modulated carrier into binary UART-framed ASCII data.
- Implemented differential-drive rover movement via PWM speed control and direction; integrated with an HTTP server/webpage over Wi-Fi for remote driving and live telemetry.
- Led project management (6 people) using Agile sprints and authored the 62-page technical report.

Discrete BJT Operational Amplifier (LTSpice)

February 2026 – March 2026

Designed an operational amplifier from individual BJTs, following a differential-gain-power stage architecture.

- Designed a differential amplifier (long-tailed pair) with a Wilson current mirror active load for high differential gain, low common mode gain, and a high output impedance for improved CMRR.
- Implemented a common-emitter amplifier stage with current mirror active load to boost voltage gain.
- Developed class AB push-pull power amplifier to increase output current and reduce crossover distortion.
- Diagnosed high-frequency instability via AC analysis of phase response (negative phase margin); stabilised with a compensation capacitor to bypass common-emitter stage at high frequencies.

Skills

Digital Design & Verification

Verilog/SystemVerilog, FPGA Synthesis (Intel Quartus), RISC CPU Architecture, RTL Verification (Icarus Verilog)

Analogue Electronics

Operational Amplifiers, Passive/Active Filters, RF Systems, PCB Design (KiCad), Arduino, Raspberry Pi

Programming

C++, Python, Ruby, C#, JavaScript, PHP, SQL

Volunteering

UK Space Design Competition

October 2025 – Present

- Provided technical advice and pastoral support to groups of 40-60 student competitors (Years 10-13) as they worked in groups to create and present engineering proposals over 1 or 2-day events.

Head Student, Kings Monkton School

September 2024 – July 2025

- Raised £1,000+ through charity events and initiatives, including managing partnerships with organisations.
- Organised school-wide community and cultural events, aimed at engaging students with real-world problems beyond academic study, such as a climate debate (70+ participants).